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Evergreening

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1 Big picture

- **Central question.** Does evergreening occur in normal credit markets, not just during banking crises or in economies with undercapitalized banks? If so, what does it do to firm borrowing, investment, survival, and aggregate productivity?
- **Paper’s definition of evergreening.** A lender with a large legacy exposure to a borrower offers the distressed borrower relatively favorable new lending terms because the new loan helps protect the old claim. The key object is a *relative subsidy*: the borrower pays better terms than it would receive from a lender that does not own the old debt.
- **Mechanism in one sentence.** A large incumbent lender would lose the old loan if the borrower defaults, so it may rationally subsidize the new loan to make default less likely.
- **Why this is not only a weak-bank story.** The paper does not require banks to be undercapitalized, hidden losses, regulatory forbearance, or gambling for resurrection. The incentive comes from concentrated ownership of legacy debt and costly borrower default.
- **Static model result.** The model creates three regions: normal lending, evergreening, and liquidation. Evergreening occurs for intermediate distress: the firm is too weak to survive at market terms, but not so weak that rescuing it is unprofitable for the incumbent lender.
- **Empirical design.** The paper uses U.S. Y-14 supervisory loan-level data from 2014:Q4–2019:Q4. For firms that borrow from multiple banks, the authors compare lending terms from banks with high versus low shares of the same firm’s debt in the same quarter.
- **Identification logic.** Firm-time fixed effects absorb borrower demand, distress, investment opportunities, and all firm-quarter shocks. The remaining variation is whether a particular bank has a larger legacy debt share in the same borrower.
- **Main bank-level evidence.** Banks with larger borrower debt shares provide more credit to distressed firms and charge them lower interest rates. Quantity rises and price falls, which is consistent with a supply subsidy.
- **Firm-level evidence.** Distressed firms with more concentrated lender structures have higher total debt growth and more investment. Thus, evergreening is not merely a reshuffling of credit across banks; it changes total firm outcomes.
- **Dynamic model contribution.** The authors embed the mechanism in a heterogeneous-firm economy with entry, exit, capital, labor, debt, default, financial frictions, and two lending regimes: dispersed lending and concentrated lending.
- **Aggregate implication.** Concentrated lending reduces interest rates, raises borrowing and capital, keeps weak firms alive, and lowers measured TFP. The paper estimates a TFP loss of roughly 0.23–0.31 percent depending on the equilibrium closure.
- **Main productivity channel.** The largest productivity loss comes from firm size under decreasing returns, not from standard static misallocation alone. Evergreening makes the average surviving firm larger.
- **Zombie-firm implication.** Subsidized firms have zombie-like characteristics: larger, older, lower productivity, higher leverage, lower interest coverage, and lower survival probabilities. But they do not necessarily pay low rates in absolute terms because they are risky.
- **Takeaway.** Evergreening can be an equilibrium feature of concentrated lending relationships. It can stabilize firms and output in the short run, but it may reduce productivity by keeping large weak firms alive.

2 Detailed section-by-section notes

2.1 Section 1: Introduction

Purpose of the section. The introduction motivates evergreening as both a crisis concern and a normal lending incentive. The authors start from the COVID-19 episode, when governments provided credit support and observers worried that banks would roll over loans to weak firms. They then ask whether the same mechanism can be observed in the United States *before* COVID-19, during a period when large banks were relatively well capitalized.

Main questions.

- Is evergreening a general feature of financial intermediation or only a pathology of crisis economies like Japan in the 1990s?
- Can the mechanism be detected using modern U.S. loan-level supervisory data?
- Does evergreening matter quantitatively for aggregate outcomes such as debt, capital, output, and TFP?

Key conceptual shift. The introduction shifts the focus from weak banks to *legacy exposure*. A bank may be healthy and still have an incentive to subsidize a distressed borrower if the bank owns enough of the borrower's old debt. The mechanism is captured by the Keynes quote: when a borrower owes enough, the lender's bargaining position weakens.

How the paper builds the argument.

1. A static model isolates the mechanism and shows why concentrated legacy lenders may offer better terms to weak firms.
2. A bank-level empirical design tests the mechanism using within-firm, within-quarter variation across banks.
3. A firm-level design tests whether lender concentration affects total debt and investment.
4. A dynamic heterogeneous-firm model measures aggregate effects and shows what subsidized firms look like.

Interpretation: The introduction is important because it defines the paper's claim narrowly. The authors are not saying all weak-firm lending is evergreening. They are saying that concentrated legacy lenders have a distinctive incentive to lend more cheaply to distressed borrowers, and this incentive is visible in both micro data and quantitative model outcomes.

2.2 Section 2: Static model overview

Purpose of the section. Section 2 develops the core mechanism in the smallest possible environment: two periods, one firm, one legacy liability, and lenders that choose the price of new debt. The goal is not realism but clarity.

Environment.

- There are two dates, $t = 0$ and $t = 1$.
- A firm enters the period with productivity z and legacy debt b .
- If the firm defaults immediately, it gets value zero.
- If the firm continues, it repays legacy debt b , borrows new funds, invests, produces, and repays the new debt.
- Lenders are risk neutral and have deep pockets.

Two lending regimes.

- **Dispersed lending economy (DLE):** new lenders do not own the legacy debt. They price only the new loan.
- **Concentrated lending economy (CLE):** one lender owns the old debt and sets new loan terms while internalizing how those terms affect default on the old claim.

Interpretation: The comparison between DLE and CLE is the heart of the paper. DLE is the market benchmark; CLE is the relationship/legacy-exposure case. The difference between them is not information, productivity, or bank health, but ownership of old claims.

2.2.1 Section 2.1: Firm problem

The firm's continuation value is

$$V(z, b; Q) = \max_{b', k' \geq 0} \{-b - k' + Qb' + \beta_f [z(k')^\alpha - b']\} \quad \text{s.t.} \quad b' \leq \theta k'. \quad (1)$$

Objects in the problem.

- z : productivity. Higher z makes continuation more attractive.
- b : legacy debt. Higher b makes continuation less attractive because it must be repaid today.
- Q : price of new debt. Higher Q means better terms and a lower interest rate.
- b' : new borrowing.
- k' : investment/capital.
- θ : collateral or borrowing constraint parameter.
- β_f : firm discount factor.

Economic meaning. The firm must pay old debt before it can continue. It can finance investment by issuing new debt at price Q . If Q is high, each unit of future repayment raises more resources today. Thus, a higher Q raises investment, borrowing, and continuation value.

Default threshold. There exists a minimum debt price $Q_{\min}(z, b)$ such that

$$\text{continue} \iff Q \geq Q_{\min}(z, b), \quad (2)$$

with

$$\frac{\partial Q_{\min}}{\partial b} > 0, \quad \frac{\partial Q_{\min}}{\partial z} < 0. \quad (3)$$

Interpretation: A more indebted firm requires better loan terms to avoid default. A more productive firm can survive at worse terms. This maps the firm's state into a default-risk threshold.

2.2.2 Proposition 1: continuation and comparative statics

Statement in words. Optimal investment k' , borrowing b' , and firm value V rise with the price of debt Q and productivity z . Firm value falls with legacy debt b . A firm defaults if the offered debt price is below $Q_{\min}(z, b)$.

Why policies rise with Q . When Q rises, new debt becomes cheaper. The firm can fund more capital with the same promised repayment. Because production has positive marginal product, investment and borrowing rise.

Why $Q_{\min}$ rises with debt. Legacy debt is a fixed repayment burden. A larger b reduces continuation value one-for-one, so the firm needs a higher debt price to make continuation worthwhile.

Why $Q_{\min}$ falls with productivity. Higher z raises the output from any chosen capital stock. The firm therefore needs less help from cheap credit to remain alive.

Interpretation: Proposition 1 gives the empirical intuition for distress: high-debt and low-productivity firms are closer to default. These are the firms for which incumbent lender incentives become most relevant.

2.2.3 Section 2.2: Dispersed lending

In the DLE, lenders price only the new debt. Since they do not own the old claim, they have no reason to subsidize the borrower. The debt price is

$$Q = \begin{cases} \beta_k, & \beta_k \geq Q_{\min}(z, b), \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

Interpretation of β_k . β_k is the competitive break-even price of new debt. It is the price atomistic lenders offer when they care only about the new loan.

Implications.

- If the firm can survive at market terms, it borrows at $Q = \beta_k$.
- If it cannot survive at market terms, it defaults.
- Among surviving firms, legacy debt does not improve lending terms.
- Weak firms do not receive special help from lenders who hold no old claims.

Interpretation: The DLE is the no-evergreening benchmark. It implies that distressed borrowers either survive at market terms or exit. There is no intermediate rescue region.

2.2.4 Section 2.3: Concentrated lending

A concentrated lender owns legacy debt b . If the firm defaults, the lender loses the old claim. The lender chooses the price of new debt subject to the firm's outside option:

$$W = \max_{Q \geq \max\{\beta_k, Q_{\min}(z, b)\}} [b + b'(z; Q)(\beta_k - Q)]. \quad (5)$$

Terms in the objective.

- b : recovery of the old debt if the firm continues.
- $b'(z; Q)(\beta_k - Q)$: profit or loss on the new debt relative to the competitive break-even price.
- If $Q > \beta_k$, the new loan is subsidized because the borrower receives better-than-market terms.

Why the bank chooses the lowest rescue price. The bank's objective is decreasing in Q when Q is above the break-even price, because a higher Q gives more subsidy on the new loan. Thus, if the bank rescues the firm, it chooses the smallest Q that prevents default: $Q = Q_{\min}(z, b)$.

2.2.5 Propositions 2 and 3: three lending regions

Let $Q_{\max}(z, b)$ be the maximum price at which the incumbent lender is willing to lend while breaking even on the combined old and new claims. The optimal policy is

$$Q^*(b, z) = \begin{cases} \beta_k, & Q_{\min}(z, b) \leq \beta_k \leq Q_{\max}(z, b), \\ Q_{\min}(z, b), & \beta_k \leq Q_{\min}(z, b) \leq Q_{\max}(z, b), \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

Region 1: normal funding. If $Q_{\min} \leq \beta_k$, the firm can survive at market terms. The concentrated lender does not need to subsidize.

Region 2: evergreening. If $\beta_k < Q_{\min} \leq Q_{\max}$, the firm would default under dispersed lending. The incumbent lender rescues it by setting $Q = Q_{\min}$. The subsidy is just large enough to keep the firm alive.

Region 3: liquidation. If $Q_{\min} > Q_{\max}$, the firm is too weak. Saving it would require too large a subsidy, so the bank lets the firm default.

Interpretation: The evergreening region is not for the strongest firms or the weakest firms. It is for intermediate-distress firms where the old claim is valuable enough to protect and the new subsidy is not too costly.

2.2.6 Section 2.4: Discussion of the static model

Potential benefit. Evergreening may save firms that are fundamentally viable but temporarily burdened by legacy debt. These firms may have positive net-present-value projects that would otherwise be lost under dispersed lending.

Potential cost. Evergreening may also preserve low-productivity firms that should exit. These firms continue using capital and labor, potentially crowding out more productive entrants.

Limitations of the static model.

- It does not determine how often firms enter the evergreening region.
- It does not model entry, exit, or aggregate reallocation.
- It does not allow firms to anticipate future rescues and choose debt strategically.
- It does not quantify the productivity cost.

Interpretation: Section 2 provides the mechanism. Sections 3–5 test and quantify it.

2.3 Section 3: Empirical analysis

Purpose of the section. The empirical section asks whether the model’s key prediction is visible in U.S. loan data: when a firm is distressed, does the bank with a larger share of the firm’s debt lend more and charge less?

2.3.1 Section 3.1: Identification approach

The baseline bank-firm regression is

$$\frac{L_{i,j,t+2} - L_{i,j,t}}{0.5(L_{i,j,t+2} + L_{i,j,t})} = \alpha_{i,t} + \beta_1 \text{DebtShare}_{i,j,t} + \beta_2 \text{DebtShare}_{i,j,t} \times \text{Distress}_{i,t} + \gamma X_{j,t} + u_{i,j,t}. \quad (7)$$

Variables.

- $L_{i,j,t}$: credit from bank j to firm i at time t .
- Dependent variable: symmetric two-quarter credit growth. This handles zeros and is bounded between -2 and 2 .
- $\text{DebtShare}_{i,j,t}$: bank j 's share of firm i 's total debt.
- $\text{Distress}_{i,t}$: indicator that firm i is in the top decile of the PD distribution.
- $\alpha_{i,t}$: firm-time fixed effects.
- $X_{j,t}$: bank controls.

Why firm-time fixed effects matter. Firm-time fixed effects absorb everything about borrower i in quarter t : loan demand, productivity, investment opportunities, distress, industry shocks, and macro shocks. Identification comes from comparing different banks lending to the same firm at the same time.

Prediction.

- For distressed firms, a high-debt-share bank should increase credit more: $\beta_2 > 0$.
- In rate regressions, a high-debt-share bank should charge lower rates: $\beta_2 < 0$.

Interpretation: The rate and quantity predictions jointly separate supply from demand. If distressed firms simply demanded more credit from high-share banks, rates should rise. Instead, higher quantities and lower rates point to a supply-side subsidy.

2.3.2 Section 3.2: Data

Main dataset. The empirical analysis uses the Federal Reserve's Y-14Q Schedule H.1 corporate loan data. These data contain detailed loan-level information for large U.S. bank holding companies.

Sample.

- Period: 2014:Q4–2019:Q4.
- The start date is chosen because PD measures become available then.
- The end date avoids COVID-19.
- The sample excludes financial and real estate firms.
- The baseline sample contains more than three million loan-facility observations and over 175,000 firms.

Distress measure. Distress is measured using banks' probability-of-default assessments. The paper uses the average borrower PD across banks. A firm is distressed if its PD is in the top decile. The top-decile cutoff is 3.89 percent, and the median firm in the top decile has a PD of about 7.8 percent.

Firm data. Private-firm balance sheets and income statements are available in Y-14. For public firms, the authors use Compustat. Firm identifiers are based on Taxpayer Identification Numbers.

Interpretation: The data are unusually well suited for this question because they observe multiple banks lending to the same firm, bank-specific debt exposure, interest rates, and internal risk assessments.

2.3.3 Section 3.3: Identifying credit-supply effects

Main bank-level result. In Table 3.1, the interaction $\text{DebtShare} \times \text{Distress}$ is positive for credit growth and negative for interest rates.

Magnitude.

- The credit-growth interaction is roughly +39 to +46 depending on controls.
- The interest-rate interaction is roughly −0.7 to −0.9 percentage points.
- The baseline interpretation is that a bank with a larger exposure to a distressed borrower expands credit more and provides better terms.

Controls and robustness in spirit. The specification includes firm-time fixed effects and bank controls. This means the result is not driven by a distressed firm borrowing more overall, nor by a common quarter shock.

Interpretation: The main empirical finding maps directly into the static model: the bank that stands to lose the most from default is the bank that has the strongest incentive to subsidize continuation.

2.3.4 Section 3.4: Comparison with zombie-firm classifications

Why this section matters. The literature often defines zombies using low interest rates, low productivity, high leverage, or weak interest coverage. The model suggests that observed low rates are not sufficient: subsidized firms can still pay high absolute rates because they are risky.

Zombie measures considered. The paper compares its PD-based distress and subsidy logic with definitions inspired by Caballero et al. (2008), Schivardi et al. (2022), Favara et al. (2022a), and high-leverage/low-productivity indicators.

Key lesson. Zombie classifications based on fundamentals such as leverage and productivity are closer to the model’s subsidized-firm concept than definitions based only on low observed interest rates.

Interpretation: The correct empirical object is a counterfactual interest-rate subsidy, not the raw level of the observed interest rate.

2.3.5 Section 3.5: Firm-level real effects

At the firm level, the specification is

$$\frac{y_{i,t+4} - y_{i,t}}{0.5(y_{i,t+4} + y_{i,t})} = \alpha_i + \tau_{t,industry,state} + \beta_1 HHI_{i,t} + \beta_2 HHI_{i,t} \times Distress_{i,t} + \beta_3 Distress_{i,t} + \gamma X_{i,t} + u_{i,t}. \quad (8)$$

Variables.

- $y_{i,t}$ is total debt or tangible assets/investment.
- $HHI_{i,t}$ measures the concentration of the firm’s lender structure.
- α_i are firm fixed effects.
- $\tau_{t,industry,state}$ are time-industry-state fixed effects.

Main firm-level finding. Distressed firms with more concentrated lenders have higher total debt growth and higher investment. The paper interprets the magnitude as economically meaningful: a distressed firm with one lender can have materially higher debt growth and investment than a similar distressed firm with dispersed lenders.

Interpretation: The bank-level result identifies a subsidy by a particular lender. The firm-level result shows that this subsidy changes firm outcomes rather than merely reallocating borrowing across lenders.

2.4 Section 4: Dynamic model

Purpose of the section. The dynamic model answers questions the static model cannot: How many firms become subsidized? How does evergreening affect entry and exit? What are the aggregate consequences?

2.4.1 Section 4.1: Setup and timing

State variables. Each firm has state

$$s = (z, b, k), \quad (9)$$

where z is productivity, b is debt, and k is capital.

Timing.

1. Productivity is realized.
2. Lenders offer a debt price $Q(s)$.
3. The entrepreneur draws repayment/default shocks.
4. The firm chooses whether to repay or default.
5. If it repays, it chooses labor, capital, and new debt.
6. If it defaults, the lender recovers part of firm value and the firm exits.
7. Entrants draw productivity and enter subject to entry costs and initial capital.

Interpretation: The model makes default probabilistic and allows the distribution of firms to evolve over time.

2.4.2 Section 4.2: Firm problem

The repayment value is

$$V^P(z, b, k; Q) = \max_{b', k', n \geq 0} \text{div} - \mathbf{1}\{\text{div} < 0\} [e_{con} + e_{slo}|\text{div}|] + \beta_f E_{z'} [\mathcal{V}(z', b', k') \mid z] \quad (10)$$

$$\text{s.t.} \quad \text{div} = zk^\alpha n^\eta - wn - k' + (1 - \delta)k + Qb' - b - c, \quad (11)$$

$$b' \leq \theta k'. \quad (12)$$

Important features.

- Production uses capital and labor with decreasing returns.
- Firms can issue equity, but negative dividends are costly.
- The collateral constraint limits debt relative to capital.
- Default is smoothed by shocks, yielding a logistic repayment probability.

The repayment probability is

$$\mathcal{P}(z, b, k; Q) = \frac{\exp(V^P(z, b, k; Q)/\kappa)}{1 + \exp(V^P(z, b, k; Q)/\kappa)}. \quad (13)$$

Interpretation: A higher debt price Q raises repayment value and survival probability. Therefore, the static model's debt-price subsidy directly carries over into default probabilities and firm dynamics.

2.4.3 Section 4.3: Alternative lending arrangements

Dispersed lending. New lenders break even on the new claim, taking into account repayment probabilities and recovery in default. They do not value legacy debt.

Concentrated lending. The incumbent lender owns the old claim and chooses Q to maximize the value of the old and new claims together:

$$W(s) = \max_{Q_r \geq Q_n(s)} \{ \mathcal{P}(s; Q_r) [b - \mathcal{B}(s; Q_r) Q_r + \beta_k E_{z'} W(z', \mathcal{B}, \mathcal{K})] + [1 - \mathcal{P}(s; Q_r)] \psi(s) \}. \quad (14)$$

Outside option. The incumbent cannot charge worse terms than the firm's new-lender option. This constraint prevents the incumbent from fully extracting the firm and focuses attention on subsidies to avoid default.

Interpretation: The CLE lender behaves like a dynamic relationship lender with a legacy balance-sheet stake. It is willing to change the new loan price because the price changes the probability of recovering old debt.

2.4.4 Section 4.4: Equilibrium closure

Constant-entry closure. The mass of entrants is fixed and wages are fixed. This resembles a sector with elastic labor supply.

Constant-labor closure. Aggregate labor supply is fixed and wages adjust. This resembles a general-equilibrium economy where additional survival of weak firms reallocates labor away from other uses.

Free entry. Entrants compare expected value with entry and initial capital costs:

$$E_\Gamma[\mathcal{V}(z', 0, \bar{k})] \geq \omega + \bar{k}. \quad (15)$$

Interpretation: The two closures help interpret aggregate outcomes. Evergreening can raise output when resources expand, but its productivity cost is clearer when total labor is fixed.

2.5 Section 5: Quantitative evaluation

2.5.1 Section 5.1: Calibration

Purpose. The calibration disciplines the dynamic model using real and financial moments. The authors calibrate the concentrated-lending economy and then compare it with dispersed lending counterfactuals.

Key externally calibrated parameters.

- Productivity persistence and volatility govern firm heterogeneity.
- Depreciation and production shares govern capital and labor demand.
- The lender discount factor governs loan pricing.
- The recovery parameter governs default losses.

Key internally calibrated parameters.

- Firm discount factor.
- Fixed operating cost.
- Logistic scale parameter for default smoothing.

- Entrant productivity distribution.
- Initial entrant capital.
- Borrowing constraint.
- Fixed equity issuance cost.

Targeted moments. The model targets leverage, debt-to-assets, investment, profit share, interest spreads, exit rates, and entrant/exit size and productivity moments from Y-14, Compustat, and firm-dynamics studies.

Interpretation: Calibration matters because the authors do not simply choose parameters to make evergreening large. They discipline the model using observable firm and credit-market moments.

2.5.2 Section 5.2: Policy functions

DLE policy functions. In the dispersed economy, more legacy debt lowers firm value and repayment probability. Debt and investment fall because outside lenders do not subsidize continuation.

CLE policy functions. In the concentrated economy, the lender raises the debt price for intermediate-debt firms. This subsidy increases repayment probability, debt, capital, and investment relative to DLE.

Where the mechanism appears. The effect is strongest for firms that are distressed but still salvageable. Very healthy firms do not need a subsidy; hopeless firms are not worth saving.

Interpretation: Policy functions are the quantitative version of Figure 2.1. They show that the evergreening region survives in the full dynamic economy.

2.5.3 Section 5.3: Aggregate effects

Main quantitative comparison. The paper compares a CLE with a DLE and studies how aggregates change when concentrated lending is allowed.

Main results.

- Interest rates fall by about 1.24 percent under constant entry and 1.13 percent under constant labor.
- Market leverage rises by about 0.60 percent and 0.54 percent.
- Capital rises by about 3.13 percent under constant entry and 1.04 percent under constant labor.
- Measured TFP falls by about 0.31 percent and 0.23 percent.
- Output rises more under constant entry because labor supply can expand; under constant labor, output barely rises.

Interpretation: Evergreening is not pure destruction. It increases debt and capital and can raise output. The cost is lower measured productivity and the survival/expansion of weaker firms.

2.5.4 Section 5.4: Aggregate productivity decomposition

The planner benchmark is

$$Y^* = M^{\frac{1}{1-\alpha-\eta}} E \left[z^{\frac{1}{1-\alpha-\eta}} \right]^{1-\alpha-\eta} K^\alpha N^\eta. \quad (16)$$

Four components. The decomposition separates aggregate output or measured TFP into:

1. average firm size;

2. selection, meaning the productivity of surviving firms;
3. static misallocation, meaning inefficient distribution of factors across firms;
4. aggregate factor quantities.

Main finding. The size component accounts for most of the productivity decline. Selection and static misallocation matter less.

Interpretation: This is a subtle result. Evergreening does not mainly reduce TFP by creating massive marginal-product dispersion. It lowers measured TFP because it changes the scale and composition of firms under decreasing returns.

2.5.5 Section 5.5: Subsidized versus non-subsidized firms

Definition of subsidized firms. A firm is subsidized if the concentrated lender charges a lower rate than the firm would face with a new/dispersed lender.

Characteristics. Subsidized firms are larger, older, less productive, more leveraged, have lower interest coverage, and have lower survival probabilities.

Why observed rates can be high. The subsidy is relative to the risk-adjusted counterfactual. A risky firm can receive a subsidy and still pay a higher observed rate than a safe firm.

Interpretation: This is why interest-rate-only zombie definitions can miss subsidized firms. The model’s relevant question is not “does the firm pay a low rate?” but “does the firm pay less than it would without the incumbent lender’s old-debt exposure?”

2.5.6 Section 5.6: Model versus data

Issue. The model initially compares extreme economies: all concentrated lending or all dispersed lending. The data contain heterogeneous lender concentration across firms.

Mixed economy. The authors also consider an economy with a mix of concentrated and dispersed lenders. This allows model-generated regressions to be compared to empirical regressions.

Result. The simulated regressions reproduce the empirical patterns: distressed firms with more concentrated lenders receive more credit and invest more.

Interpretation: This closes the loop: the same mechanism explains the micro evidence and produces aggregate productivity consequences.

2.6 Section 6: Conclusion

Main conclusion. Evergreening is a normal consequence of concentrated lending relationships with costly default. It need not reflect hidden losses, weak banks, or crisis forbearance.

Empirical conclusion. U.S. banks with larger debt shares in distressed firms provide more credit at lower rates, even in a relatively healthy pre-COVID period.

Quantitative conclusion. Evergreening lowers interest rates and raises debt/capital but reduces measured productivity by keeping weaker firms alive and larger.

Interpretation: The policy implication is balanced. Evergreening can stabilize distressed borrowers but may create longer-run productivity costs. Detecting it requires lender-level exposure and risk-adjusted counterfactuals, not simple low-rate zombie labels.

3 Model and empirical equations cheat sheet

3.1 Firm continuation value in the static model

$$V(z, b; Q) = \max_{b', k' \geq 0} \{-b - k' + Qb' + \beta_f[z(k')^\alpha - b']\} \quad \text{s.t.} \quad b' \leq \theta k'. \quad (17)$$

Interpretation: The firm survives only if the new debt price is high enough to offset the burden of legacy debt.

3.2 Dispersed lending debt price

$$Q = \beta_k \quad \text{if } \beta_k \geq Q_{\min}(z, b), \quad Q = 0 \text{ otherwise.} \quad (18)$$

Interpretation: Dispersed lenders care only about the new loan, so there is no subsidy.

3.3 Concentrated lender objective

$$W = \max_{Q \geq \max\{\beta_k, Q_{\min}\}} [b + b'(z; Q)(\beta_k - Q)]. \quad (19)$$

Interpretation: The lender may lose money on the new loan to recover the old loan.

3.4 Bank-level empirical regression

$$\Delta L_{ijt} = \alpha_{it} + \beta_1 \text{DebtShare}_{ijt} + \beta_2 \text{DebtShare}_{ijt} \times \text{Distress}_{it} + \gamma X_{jt} + u_{ijt}. \quad (20)$$

Interpretation: A positive β_2 for credit growth and a negative β_2 for rates is the signature of evergreening.

3.5 Firm-level empirical regression

$$\Delta y_{it} = \alpha_i + \tau_{t, \text{industry, state}} + \beta_1 \text{HHI}_{it} + \beta_2 \text{HHI}_{it} \times \text{Distress}_{it} + \beta_3 \text{Distress}_{it} + \gamma X_{it} + u_{it}. \quad (21)$$

Interpretation: This tests whether concentrated lending changes total firm debt and investment.

3.6 Dynamic repayment probability

$$\mathcal{P}(z, b, k; Q) = \frac{\exp(V^P(z, b, k; Q)/\kappa)}{1 + \exp(V^P(z, b, k; Q)/\kappa)}. \quad (22)$$

Interpretation: Better loan terms increase continuation value and therefore repayment probability.

4 Tables, figures, and original-paper excerpts

4.1 Figure 2.1: concentrated lending mechanism

Read:

- The horizontal axis is legacy debt b .
- The key thresholds are β_k , $Q_{\min}$, and $Q_{\max}$.
- Low b : the firm survives at market terms.
- Intermediate b : the firm needs a subsidy, and the incumbent lender provides one.
- High b : the subsidy needed is too large, so the bank liquidates the firm.

Economic message: The figure is the simplest visual summary of the mechanism: evergreening is an intermediate-distress region created by a lender’s exposure to old debt.

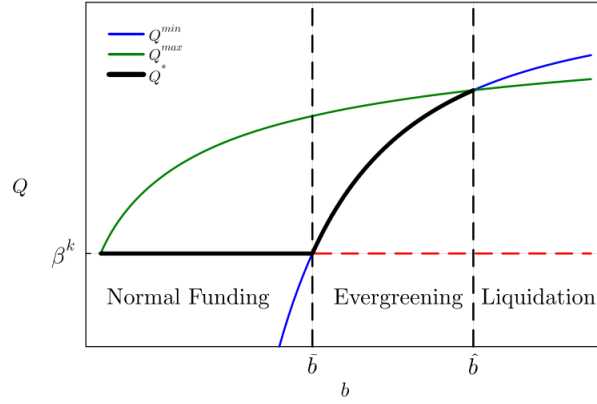


Fig. 2.1. Concentrated lending economy. **Notes:** Equilibrium allocation as a function of b , for a given z . The solid blue line is $Q^{\min}(z, b)$, the solid green line is $Q^{\max}(z, b)$, the dashed red line is β^k , and the black line is the optimal policy Q^* . (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Proposition 3 states that, as long as legacy debt is positive, $b > 0$, the bank is willing to offer better terms than those in the competitive market to the firm. Offering more favorable lending conditions allows the bank to recover b by preventing the firm from defaulting. The optimal price of debt Q^* consists of three regions, illustrated in Fig. 2.1 for a fixed z . First, as long as $Q^{\min}(z, b) < \beta^k$, the bank can of-

4.2 Tables 3.1 and 3.2: bank-level and firm-level empirical evidence

Read: Table 3.1:

- The interaction $\text{DebtShare} \times \text{Distress}$ is positive in credit-growth regressions.
- The same interaction is negative in interest-rate regressions.
- Quantity rises and price falls for high-share banks lending to distressed firms.
- This is the core empirical signature of a supply-side evergreening subsidy.

Read: Table 3.2:

- At the firm level, lender concentration matters for total outcomes.
- Higher HHI among distressed firms raises total debt growth and investment.
- This indicates that bank-level subsidies are not fully offset by reductions elsewhere.

Economic message: The bank-level evidence identifies the mechanism; the firm-level evidence shows real effects.

Table 3.1

Credit supply to distressed firms.

	Δ Credit			Δ Interest Rate		
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
Debt Share	-21.88** (8.24)	-17.48** (8.58)	-22.37*** (7.84)	0.18*** (0.05)	0.11 (0.07)	0.12* (0.06)
Debt Share $\times$ Distress	43.60*** (9.49)	38.56*** (10.50)	44.95*** (12.84)	-0.91*** (0.33)	-0.71** (0.33)	-0.72** (0.32)
Fixed Effects	✓	✓	✓	✓	✓	✓
Firm $\times$ Time	✓	✓	✓	✓	✓	✓
Firm $\times$ Time $\times$ Pur.	✓	✓	✓	✓	✓	✓
Bank Controls	✓	✓	✓	✓	✓	✓
R-squared	0.58	0.6	0.63	0.74	0.74	0.79
Observations	6,647	5,729	8,576	8,407	5,561	8,238
w/ Distress = 1	539	397	531	528	386	520
Number of Firms	887	642	884	867	621	864
Number of Banks	36	34	34	36	34	34

Notes: Estimation results for regression (3.1) multiplied by 100. All specifications include firm-time fixed effects that additionally vary by the loan purpose in columns (ii) and (v). Columns (iii) and (vi) include bank-time fixed effects and the remaining columns include various bank controls: bank size (natural log of assets), return on assets (net income/assets), deposit share (total deposits/assets), loan share (loans/assets), leverage (liabilities/assets), Tier 1 capital buffer (ratio minus requirement), banks' income gap, and the ratio of unused credit lines to assets. Standard errors in parentheses are two-way clustered by bank and firm. Sample: 2014:Q4 - 2019:Q4. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

at time i , and the interaction of this variable with a firm-specific indicator $Distress_{i,t}$ that equals one if firm i is in financial distress at time t and zero otherwise. We classify a firm as financially distressed if the average PD across all banks that lend to this firm, denoted by $\overline{PD}_{i,t}$, falls within the top decile of the unconditional distribution of $\overline{PD}_{i,t}$ across all firms within our sample. We choose this definition for the following three reasons. First, we use firms' PDs since those can be understood as *sufficient statistics* to measure firm financial distress and because they directly relate to our theory, which is concerned with firms' distance to default. We use a binary indicator variable, as opposed to the continuous variable $\overline{PD}_{i,t}$, since our theory shows that the relation between firm distress and lender's willingness to evergreen loans is inherently nonlinear (as visible in Appendix Fig. A.3), and the indicator $Distress_{i,t}$ is a simple approximation of this relation. Second, we average PDs

$r_{i,t}$ denotes the interest rate associated with the credit agreement between firm i and bank j .¹⁰ If our results represent demand shifts, the estimated coefficients from the interest rate regressions should have the same signs as the corresponding ones from the credit regressions. In contrast, if we capture credit supply effects, they should have the opposite sign.

Last, we exclude credit lines and focus on term loans only. That is because credit movements for credit lines largely represent demand changes. Such contracts provide borrowers with the possibility to flexibly draw and repay credit subject to a predetermined limit and at a fixed spread (Greenwald et al., 2021).¹¹ As shown below, our key results remain if we include credit lines, but the interest rate responses indicate that the findings may be driven by demand shifts.

The estimation results for regressions (3.1) are reported in Ta-

Table 3.2

Credit supply to distressed firms - firm level.

	Δ Total Debt		Investment		Δ Ave. Interest Rate	
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
HHI	33.71*** (8.27)	32.79*** (8.30)	11.82*** (3.86)	11.81*** (3.92)	-2.21*** (0.42)	-2.22*** (0.42)
HHI $\times$ Distress	13.34*** (4.54)	19.49*** (5.41)	6.88** (3.49)	7.55** (3.85)	-0.78 (0.82)	-0.82 (0.89)
Distress	4.38*** (1.38)	-2.24*** (1.83)	-2.56*** (0.71)	-2.84*** (0.86)	0.04 (0.10)	-0.01 (0.13)
Fixed Effects	✓	✓	✓	✓	✓	✓
Firm	✓	✓	✓	✓	✓	✓
Time $\times$ Industry $\times$ State	✓	✓	✓	✓	✓	✓
Firm Controls $\times$ Distress	✓	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓	✓
R-squared	0.56	0.56	0.58	0.58	0.44	0.44
Observations	60,636	60,636	71,854	71,854	55,222	55,222
w/ Distress = 1	5,211	5,211	6,195	6,195	4,896	4,896
Number of Firms	14,400	14,400	17,063	17,063	13,021	13,021
Number of Banks	37	37	37	37	37	37

Notes: Estimation results for regression (3.2) multiplied by 100, where $y_{i,t}$ is either total firm debt in columns (i) and (ii) or tangible assets in columns (iii) and (iv). Columns (v) and (vi) estimate regression (3.2) using the change in the average interest rate $r_{i,t+1} - r_{i,t}$ as a dependent variable. All specifications include firm fixed effects, time-industry-state fixed effects where an industry is classified using two-digit NAICS codes, the ratio of observed debt to total debt, and various firm controls: cash holdings, tangible assets, liabilities, debt, net income (all scaled by total assets), and firm size (natural logarithm of total assets). Columns (i), (iv), and (vi) include interactions of each of the demanded firm controls with the distress indicator. Standard errors in parentheses are two-way clustered by firm and the bank with the largest debt-share. Sample: 2014:Q4 - 2019:Q4. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 1: Original paper Table 3.1: credit supply to distressed firms.

Table 2: Original paper Table 3.2: firm-level debt and investment effects.

4.3 Equations (3.1) and (3.2): empirical specifications

Read:

- Equation (3.1) is the bank-firm credit-supply test.
- Equation (3.2) is the firm-level real-effects test.
- Firm-time fixed effects in equation (3.1) are central because they absorb borrower demand.
- Firm fixed effects and time-industry-state fixed effects in equation (3.2) discipline cross-firm comparisons.

Economic message: The two equations together move from credit supply to real outcomes.

we estimate regressions of the form

$$\frac{L_{i,j,t+2} - L_{i,j,t}}{0.5 \cdot (L_{i,j,t+2} + L_{i,j,t})} = \alpha_{i,t} + \beta_1 \text{Debt-Share}_{i,j,t} + \beta_2 \text{Debt-Share}_{i,j,t} \times \text{Distress}_{i,t} + \gamma X_{j,t} + u_{i,j,t} \quad (3.1)$$

where $L_{i,j,t}$ is the aggregated amount of credit between a bank and a firm at time t , and the dependent variable measures percentage changes in credit over two quarters. Specifically, we use the symmetric growth rate as an approximation of a percentage change, which allows for possible zero observations at time t and is bounded in the range $[-2, 2]$, reducing the potential influence of outliers. We include firm-time fixed effects $\alpha_{i,t}$ into our regressions, which restrict the sample to firms that borrow from multiple banks. The fixed effects control for credit demand if firms have a common demand across their lenders, and we discuss possible violations of this assumption below.

The main regressors are $\text{Debt-Share}_{i,j,t}$, defined as the ratio of outstanding credit $L_{i,j,t}$ between firm i and bank j to total firm debt $\text{Debt}_{i,t}$

the various (demeaned) firm controls and the distress indicator. We find $\beta_1 > 0$ for both outcome variables, which are statistically different from zero at the one percent confidence level. These results can be explained by the fact that—relative to other nondistressed firms—the ones with more concentrated debt positions are potentially younger firms that are

zombie firm as one with negative sales growth over the previous three years, a leverage ratio above the median across all firms, and an interest coverage ratio below one.

¹⁵ For leverage, we use the 90th percentile across all firms for a particular period as a cutoff. For productivity, we use the 10th percentile of firms' return on assets for a particular period as a cutoff.

¹⁶ The mass of firms with low PDs is relatively large for zombies under all three measures, with more than half of the firms having a PD of 2.8 percent or less. That makes it unlikely that these are all firms that are saved whenever they experience some distress, so that their lenders receive low PDs with such

Figure 1: Original paper equation (3.1): bank-level identification.

Figure 2: Original paper equation (3.2): firm-level specification.

4.4 Dynamic model problem

Read:

- The excerpt contains the dynamic repayment value, dividend equation, borrowing constraint, and repayment probability.
- The model includes production, labor choice, capital accumulation, debt issuance, equity issuance costs, and default.
- The logistic repayment probability smooths default decisions and makes default risk continuous.

Economic message: The dynamic model is the quantitative bridge from the lender-level subsidy to aggregate productivity.

THE FIRM PROBLEM

Firms have access to a decreasing returns-to-scale production technology with the production function given by $z k^\alpha n^\eta$, where z is current productivity, k is current capital, and n is labor. The capital share is α , and the labor share is η . The firm hires labor at wage w and invests in new capital k' at a constant unit cost. Capital depreciates at rate δ . Additionally, the firm pays a fixed cost of operation equal to c . The value of repaying conditional on today's state $s = (z, k, b)$ and the offered contract Q is given by

$$V^r(z, b, k; Q) = \max_{b' \in \mathbb{R}} d|v - \mathbb{I}[d|v < 0][e_{\text{con}} + e_{\text{de}}|d|v|] + \beta^r \mathbb{E}_r[V(z', b', k')|z] \quad (4.1)$$

$$\text{s.t. } d|v = z k^\alpha n^\eta - w n - k' + (1 - \delta)k + Qb' - b - c, \quad (4.2)$$

$$b' \leq \theta k' \quad (4.3)$$

The value of repayment is equal to current dividends $d|v$ plus the continuation value, which is explained below. The firm is also subject to equity issuance costs, with a fixed cost component e_{con} and a linear cost scaled by e_{de} . Equation (4.2) defines the firm dividend: the value of production, minus the wage bill, minus the new investment net of undepreciated capital, plus new borrowings, minus debt repayments, and minus the fixed cost. Equation (4.3) is a borrowing constraint as in the static model. We refer to the policy functions that solve this problem as $B(z, b, k; Q)$ and $K(z, b, k; Q)$, and the optimal labor choice results from a simple static problem.

The firm chooses how much to borrow b' for an offered price of debt Q that is taken as given. In this sort of environment with defaultable debt, a borrowing constraint is required for an equilibrium to be well-defined (see, for example, Ayres et al., 2018). However, due to precautionary behavior arising from the interaction between the expectation of future shocks and equity issuance costs, the borrowing constraint may not necessarily bind.

The firm's value before deciding repayment, after receiving an offer Q , and upon realizing the additive shocks z' and $k'^{\alpha, \beta}$ can be written as $V_f(z, b, k, e^{\beta^r} z', Q) = \max_{b'} \{V^r(z, b, k; Q) + e^{\beta^r} 0 + e^{\beta^r} 0\}$, where

THE LENDER PROBLEM
The price of debt Q is determined by a free-entry condition for lenders. Given $s = (z, b, k)$, we use the notation $Q^*(s)$ to refer to the dispersed lending equilibrium price, which is the price that satisfies the following zero expected-discounted profit condition

$$0 = -Q^* B(s; Q^*) + \beta^l \mathbb{E}_l[P(z', B(s; Q^*), K(s; Q^*)) B(s; Q^*)] + [1 - P(z', B(s; Q^*), K(s; Q^*))] \psi(z', B(s; Q^*), K(s; Q^*)) \quad (4.6)$$

where ψ is the recovery value in case of default. This value is given by a fraction ψ_1 of the revenue generated by producing one last period and liquidating the undepreciated stock of capital, i.e. $\psi(z, b, k) \equiv \psi_1 [\max_z z k^\alpha n^\eta - w n + (1 - \delta)k - c]$. The expression for the price resembles the one used in models of sovereign default, with the difference that we have to take into account the firm choices for capital and debt, $K(s; Q)$ and $B(s; Q)$, which are determined after Q^* is offered. Note that we assume that lenders have a discount factor larger than that of the firm, $\beta^l > \beta^r$. This assumption ensures that firms never fully escape their constraints, even in the long run (Rampini and Viswanathan, 2013). It is also similar to the existence of a tax advantage of debt as it distorts firms' choice of capital structure towards debt (Kiyotaki and Moore, 1997; Li et al., 2016).

Concentrated lending economy (CLE) The second type of credit market we study is one where lenders internalize the firm choices and the possibility of default on current claims b when choosing lending terms. Consequently, such concentrated lenders may offer a different Q that we denote by $Q^c(s)$. The market power that an existing lender can exercise is limited since a large mass of potential lenders stands ready to start a new relationship with a firm.²⁰ The problem of a lender that has lent b in the previous period to a firm with current capital k and productivity z is

$$W(s) = \max_{Q \geq Q^*(s)} P(s; Q^c) [b - B(s; Q^c) Q^c] + \beta^l \mathbb{E}_l[W(z', B(s; Q^c), K(s; Q^c))|z] + [1 - P(s; Q^c)] \psi(s) \quad (4.7)$$

where $Q^c(s)$ is the price offered by new lenders. Given this free entry

4.5 Tables 5.1 and 5.2: calibration and fit

Read:

- Table 5.1 lists external and internal parameters.
- Table 5.2 compares model moments to data moments.

- The model is disciplined by leverage, debt, investment, profit shares, spreads, exit, and entrant/exit statistics.

Economic message: The quantitative exercise is credible only if the model can match both financial and real firm moments before comparing lending regimes.

Table 5.1
Model parameters and values.

Parameter	Description	Value	Source/Reason
ω	Cost of entry	1.184	Normalize $w = 1$
ρ_z	TFP persistence	0.767	Gomes (2001), Gourio and Miao (2010)
σ_π	TFP volatility	0.110	Gomes (2001), Gourio and Miao (2010)
c_{issue}	Equity issuance cost	0.200	Hennessy and Whited (2007)
δ	Depreciation rate	0.100	Aggregate investment/capital of 10%
α	Production, capital share	0.320	Profit share of 16%
η	Production, labor share	0.480	Profit share of 16%
β^k	Lender discount rate	0.970	Real rate of 3%
ψ_1	Recovery value	0.350	Kermani and Ma (2020)
β^l	Borrower discount factor	0.884	Internally calibrated
$\mathbf{c}$	Fixed cost	0.055	Internally calibrated
κ	Logistic distr., scale	0.225	Internally calibrated
z	TFP distr. for entrants	1.147	Internally calibrated
k	Initial capital	0.805	Internally calibrated
θ	Constraint parameter	1.040	Internally calibrated
c_{out}	Fixed cost of issuing equity	0.010	Internally calibrated

Table 5.2
Data moments and model fit.

Moment	Source	Data	Model
Market leverage (median)	Y-14/Compustat	0.63/0.57	0.59
Debt over fixed assets (median)	Y-14/Compustat	1.09/1.20	1.04
Investment rate (aggregate)	Y-14/Compustat	0.104/0.14	0.117
Profit share (aggregate)	Y-14	0.16	0.176
Interest rate spread (median)	Y-14	3.46%	4.47%
Exit rate	Hopenhayn et al. (2022)	9.0%	8.8%
Size at entry (relative to mean)	Lee and Mukoyama (2015)	0.60	0.58

4.6 Figure 5.1 and Table 5.3: policy functions and aggregates

Read: Figure 5.1:

- The concentrated lender offers better prices to intermediate-distress firms.
- These better prices raise repayment probability, borrowing, and capital.
- Very low-debt firms and very high-debt firms are less affected.

Read: Table 5.3:

- Concentrated lending lowers interest rates.
- It raises debt and capital.
- It lowers measured TFP.
- Output rises more when labor supply/entry is elastic than when labor is fixed.

Economic message: Evergreening is expansionary in credit quantities but contractionary for productivity.

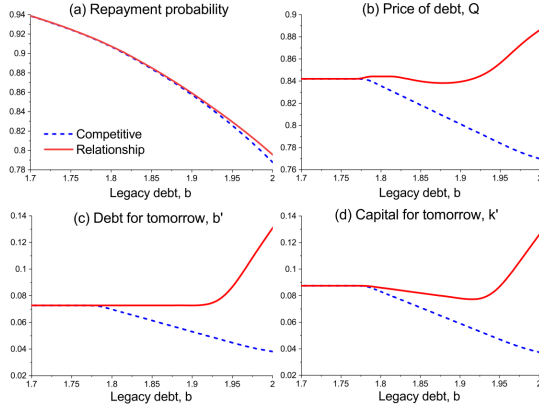


Fig. 5.1. Comparison of policy functions. **Notes:** Policy functions and values for a firm with the same set of $(z = 0.6, k = 2)$, as a function of b , dispersed lending (blue, dashed) vs. concentrated lending (red, solid) economies.

Figure 3: Original paper Figure 5.1: policy functions.

Table 5.3

Impact of introducing concentrated lending.

	Δ % with const. entry	Δ % with const. labor
<i>Firm level (Averages)</i>		
Market Leverage	0.60	0.54
Interest rate	-1.24	-1.13
Size	2.34	1.99
Productivity	-0.04	-0.02
Exit rate	-0.70	-0.17
<i>Aggregates</i>		
Debt	3.13	1.04
Capital	3.13	1.04
Labor	2.14	0.00
Output	2.14	0.10
Wage	0.00	0.10
Measured TFP	-0.31	-0.23
Number of firms	0.77	-0.94

Notes: Size is measured in terms of capital. Measured TFP is given by $Y/(K^\alpha N^{1-\alpha})$.

rium concepts differ with respect to the number of firms. With constant entry, the number of firms increases as more firms survive, and the measure of entrants is constant. In contrast, with constant labor, the number of firms declines slightly because firms are larger, which implies that fewer resources are available for new entrants, leading to a drop in firm entry.

5.4. Aggregate productivity in the CLE and the DLE

Our results suggest that the lending regime affects the average size and profitability of incumbent firms, both of which could affect aggregate productivity. We decompose aggregate productivity under each lending regime into three separate terms: static misallocation in the

Table 3: Original paper Table 5.3: aggregate effects.

4.7 Proposition 4 and equation (5.1): productivity decomposition

Read:

- Proposition 4 gives the efficient allocation benchmark under fixed mass of firms and aggregate inputs.
- Equation (5.1) decomposes output ratios into size, selection, static misallocation, and factor quantities.
- The decomposition helps identify why evergreening lowers measured TFP.

Economic message: The main productivity loss is not simply “bad allocation of capital” in the narrow static sense; it is also about firm scale and survival under decreasing returns.

Proposition 4. In an economy where a planner can freely reallocate capital and labor across firms to maximize production, for a given mass of firms, aggregate production is given by $Y^* = M^{1-\alpha-\eta} \mathbb{E}[z^{\frac{1}{1-\alpha-\eta}}]^{1-\alpha-\eta} K^\alpha N^\eta$, where $K \equiv \int_s k(s) d\lambda(s)$, $N \equiv \int_s n(s) d\lambda(s)$ are the aggregate stocks of capital and labor, respectively. Proof: See Appendix D.1.

As a direct corollary we can write output in the decentralized economy as

$$Y = \underbrace{\left(\frac{1}{S}\right)^{1-\alpha-\eta}}_{\text{avg. firm size}} \times \underbrace{\mathbb{E}[z^{\frac{1}{1-\alpha-\eta}}]^{1-\alpha-\eta}}_{\text{selection}} \times \underbrace{\frac{Y}{Y^*}}_{\text{static misallocation}} \times \underbrace{K^\alpha N^{1-\alpha}}_{\text{factor qtys.}},$$

where $S \equiv N/M$ is the average firm size. The first three terms correspond to measured TFP, $MTFP \equiv Y/K^\alpha N^{1-\alpha}$. MTFP depends on three components: the first term is average firm size. This term appears since firms operate with decreasing returns to scale technology: an economy with more and/or smaller firms has higher MTFP, everything else constant. The second term represents selection, or dynamic misallocation: an economy with more productive incumbents on average has higher MTFP, everything else constant. The final term represents static misallocation in the sense of Hsieh and Klenow (2009). It is equal to 1 in an economy where a constant amount of factor inputs are distributed to

Figure 4: Original paper Proposition 4 and planner benchmark.

$$Y_j = \underbrace{\left(\frac{1}{S_j}\right)^{1-\alpha-\eta}}_{\text{avg. firm size}} \times \underbrace{\mathbb{E}_j[z^{\frac{1}{1-\alpha-\eta}}]^{1-\alpha-\eta}}_{\text{selection}} \times \underbrace{\frac{Y_j}{Y_j^*}}_{\text{static misallocation}} \times \underbrace{K_j^\alpha N_j^{1-\alpha}}_{\text{factor qtys.}} \quad (5.1)$$

Table 5.4 reports the results of the decomposition of MTFP for the CLE vs. the DLE with constant entry or constant labor. MTFP is lower in the CLE in both cases: the decomposition attributes most of this drop to the size component, as firms are on average larger in the CLE. There is also a small negative contribution from selection, as firm productivity is also lower on average in the CLE. Finally, static misallocation is worse in the CLE, suggesting that subsidized lending also worsens static efficiency. However, it accounts for only around 10% of MTFP losses, with the bulk arising from firm size. This suggests that traditional measures of static misallocation, such as the standard deviation of MPK, may not be informative regarding productivity losses generated by lending ar-

Figure 5: Original paper equation (5.1): decomposition.

4.8 Tables 5.4 and 5.5: TFP losses and subsidized firms

Read: Table 5.4:

- MTFP falls in both constant-entry and constant-labor comparisons.
- The size component explains most of the decline.
- Static misallocation and selection are smaller channels.

Read: Table 5.5:

- Subsidized firms have more capital, higher leverage, lower productivity, lower interest coverage, and lower survival probability.
- They can still pay high observed interest rates because they are riskier.

Economic message: Subsidized firms resemble zombies in fundamentals, but the observed interest rate is not enough to identify them.

Table 5.4
MTFP decomposition: CLE vs. DLE.

Ratio	% Δ DLE constant entry to CLE	% Δ DLE constant labor to CLE
MTFP	-0.309	-0.227
Size	-0.270	-0.188
Selection	-0.008	-0.004
Static Misallocation	-0.032	-0.035

Table 5.5
Subsidized vs. non-subsidized Firms in the CLE (medians).

	Non-subsidized	Subsidized	Δ %
Capital	0.75	1.72	128.5
Productivity	1.02	0.94	-8.0
Output	0.41	0.60	46.1
Payouts/assets	0.05	-0.01	-114.4
Market leverage	0.53	0.80	50.6
Interest rate	7.75	10.02	29.2
Probability of survival	0.96	0.89	-7.6
Interest-coverage ratio	1.67	0.45	-73.1
Age	7.87	10.17	29.2

$$\frac{Y_i}{Y_i^*} = \left(\frac{1/S_i}{1/S_i^*} \right)^{1-\alpha-\eta} \times \left(\frac{E_i[z^{\frac{1}{1-\alpha-\eta}}]}{E_i^*[z^{\frac{1}{1-\alpha-\eta}}]} \right)^{1-\alpha-\eta} \times \left(\frac{Y_i/Y_i^*}{Y_i/Y_i^*} \right) \times \left(\frac{K_i^\alpha N_i^{1-\alpha}}{K_i^{\alpha*} N_i^{1-\alpha*}} \right).$$

as those offered by a new lender without evergreening incentives—a counterfactual that cannot be observed in the data.

Subsidized vs. zombie firms While there is a large empirical literature that attempts to classify zombie firms, there is no single definition of what constitutes one, and a wide range of classification methodologies have been proposed in the literature. We focus on the measure by Favara et al. (2022a) (FMP), who quantify the number of zombie firms in the U.S. using a similar dataset to ours. They classify a firm as a zombie if it satisfies the following three conditions: (i) leverage above the median, (ii) an interest coverage ratio below 1, and (iii) average negative sales growth over the past 3 years. Given our calibration, we find that 5.7% of firms satisfy this definition in the stationary equilibrium with concentrated lending. This is consistent with the estimates of FMP, who find a zombie firm share of 5.6%-5.7% between 2017 and 2019. This is a completely understated and relevant moment; thus, we

5 Short summary

- Evergreening arises when a lender with old exposure subsidizes new lending to prevent borrower default.
- The static model has three regions: normal funding, evergreening, and liquidation.
- Evergreening occurs for intermediate-distress firms.
- The paper tests the mechanism with Y-14 loan-level data from 2014:Q4–2019:Q4.
- Firm-time fixed effects allow the authors to compare banks lending to the same firm in the same quarter.
- High-debt-share banks lend more to distressed firms and charge lower rates.
- Distressed firms with concentrated lenders have higher total debt growth and investment.
- Interest-rate-only zombie definitions are incomplete because subsidized firms can pay high rates in absolute terms.
- The dynamic model embeds the mechanism into a heterogeneous-firm economy.
- Concentrated lending lowers rates, raises debt and capital, and lowers measured TFP.
- The TFP cost is mostly due to firm size under decreasing returns.
- Subsidized firms are larger, older, less productive, more leveraged, and have lower interest coverage.
- The paper’s main message is that evergreening can exist even in healthy banking systems and normal macroeconomic conditions.

6 Conclusion

6.1 Core conclusion

Faria-e-Castro, Paul, and Sánchez show that evergreening can arise from concentrated legacy lending. A lender that owns a large share of a distressed firm’s old debt may rationally offer favorable new terms to preserve the value of the old claim. This behavior appears in U.S. supervisory loan-level data and has meaningful quantitative consequences in a dynamic firm model.

6.2 Economic intuition

A new lender asks only whether the new loan breaks even. An incumbent lender asks whether the new loan helps recover the old loan. This difference makes weak firms with large legacy exposure potentially attractive to rescue. The rescue is not charity: it is a way of maximizing the incumbent lender’s combined claim value.

6.3 Policy and measurement takeaway

Evergreening can support firms and investment in the short run, but it can reduce productivity by keeping weak firms large and alive. Identifying evergreening requires counterfactual thinking: compare the observed loan terms with the terms the firm would receive from a lender without legacy exposure. Raw low interest rates alone are not enough.